

Filtros de *Blueprints*

La idea es:

- Tener una interfaz BlueprintsFilter
- Implementaciones:
 - RedundancyFilter
 - UndersamplingFilter
- Activarlas usando perfiles:
 - redundancy
 - undersampling
- Inyectar el filtro activo en el servicio
- Aplicarlo antes de devolver el blueprint

```
8  /**
9   * Default filter: returns the blueprint unchanged.
10  * This matches the baseline behavior of the reference lab before
11  */
12  @Component
13  @Profile("!redundancy & !undersampling")
14  public class IdentityFilter implements BlueprintsFilter {
15
16      @Override
17      public Blueprint apply(Blueprint bp) {
18          return bp;
19      }
20  }
```

Se activa cuando ninguno de los perfiles anteriores está activo

Aplicar el filtro en el Service

```

13 import java.util.stream.Collectors;
14
15 @Service
16 public class BlueprintsServices {
17
18     private final BlueprintPersistence persistence;
19     private final BlueprintsFilter filter;
20
21     @Autowired
22     public BlueprintsServices(BlueprintPersistence persistence,
23                               BlueprintsFilter filter) {
24         this.persistence = persistence;
25         this.filter = filter;
26     }
27
28     public void addNewBlueprint(Blueprint bp)
29         throws BlueprintPersistenceException {
30
31         persistence.saveBlueprint(bp);
32     }
33
34     public Set<Blueprint> getAllBlueprints() {
35
36         return persistence.getAllBlueprints()
37             .stream()
38             .map(filter::apply)
39             .collect(Collectors.toSet());
40     }
41
42     public Set<Blueprint> getBlueprintsByAuthor(String author)
43         throws BlueprintNotFoundException {
44
45         return persistence.getBlueprintsByAuthor(author)
46             .stream()
47             .map(filter::apply)
48             .collect(Collectors.toSet());
49     }
50
51     public Blueprint getBlueprint(String author, String name)
52         throws BlueprintNotFoundException {
53
54         return filter.apply(
55             persistence.getBlueprint(author, name)
56         );
57     }
58
59     public void addPoint(String author, String name, int x, int y)
60         throws BlueprintNotFoundException {
61
62         persistence.addPoint(author, name, x, y);
63     }
64 }

```

Si no se aplica el filtro en el servicio, el perfil no hace nada.

Activar perfiles

```
1 spring.mvc.pathmatch.matching-strategy=ant_path_matcher
2 spring.datasource.url=jdbc:postgresql://localhost:5432/blueprintsdb
3 spring.datasource.username=postgres
4 spring.datasource.password=postgres
5 spring.datasource.driver-class-name=org.postgresql.Driver
6
7 spring.sql.init.mode=never
8 spring.profiles.active=redundancy
9 # spring.profiles.active=undersampling
```

Pruebas

```
{
  "author": "juanito",
  "name": "plano1",
  "points": [
    {"x": 10, "y": 10},
    {"x": 10, "y": 10},
    {"x": 20, "y": 20},
    {"x": 30, "y": 30},
    {"x": 30, "y": 30},
    {"x": 40, "y": 40}
  ]
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8080/api/v1/blueprints' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "author": "juanito",
    "name": "plano1",
    "points": [
      {"x": 10, "y": 10},
      {"x": 10, "y": 10},
      {"x": 20, "y": 20},
      {"x": 30, "y": 30},
      {"x": 30, "y": 30},
      {"x": 40, "y": 40}
    ]
  }'
```

Request URL

http://localhost:8080/api/v1/blueprints

Server response

Code

Details

201

Response body

```
{
  "code": 201,
  "message": "Blueprint created",
  "data": null
}
```

Download

Response headers

```
connection: keep-alive
content-type: application/json
```

redundancy

```
    }
  },
  {
    "author": "juanito",
    "name": "plano1",
    "points": [
      {
        "x": 10,
        "y": 10
      },
      {
        "x": 20,
        "y": 20
      },
      {
        "x": 30,
        "y": 30
      },
      {
        "x": 40,
        "y": 40
      }
    ]
  },
  {
    "author": "Andres",
    "name": "plano3",
    "points": [

```

Eliminó duplicados consecutivos

Conserva el orden

undersampling

```
    {
      "author": "juanito",
      "name": "plano1",
      "points": [
        {
          "x": 10,
          "y": 10
        },
        {
          "x": 20,
          "y": 20
        },
        {
          "x": 30,
          "y": 30
        }
      ]
    },
    {
      "author": "Andres",

```

Conserva 1 de cada 2 puntos

Toma índices 0,2,4